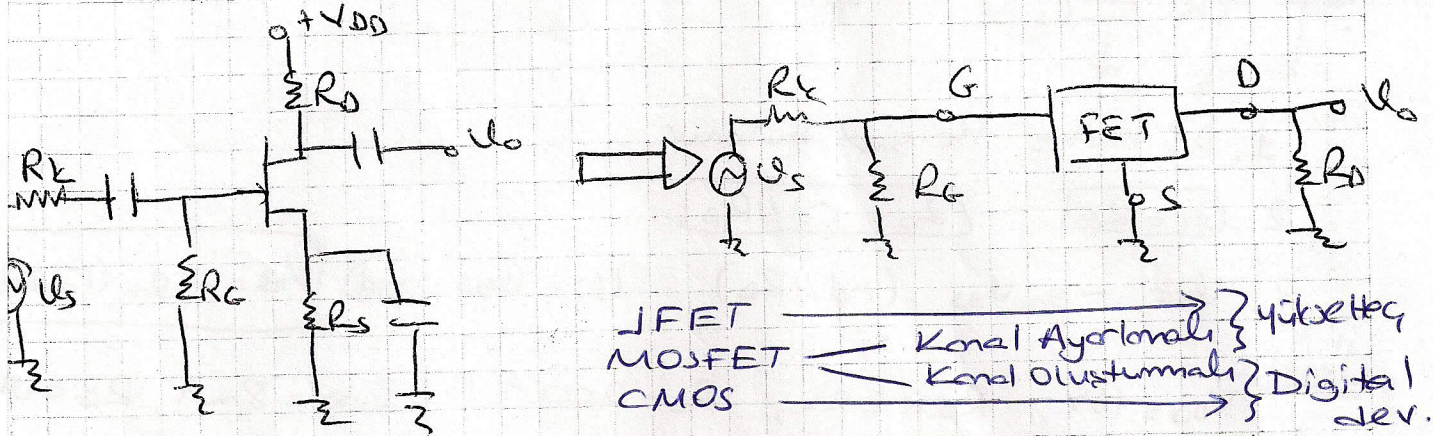


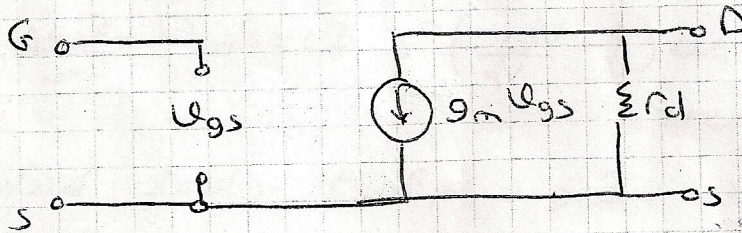
BÖLÜM-3

FET KÜÇÜK SINYAL ANALİZİ

FET'in giriş direnci yüksektir. Bu nedenle yüksek giriş direnci olan bir yükselteç tasarlamak ise tercih edilir. FET daha az gürültü üretir.



JFET ve Kanal Ayrıklı MOSFET'in AC Modeli: Kullılır.



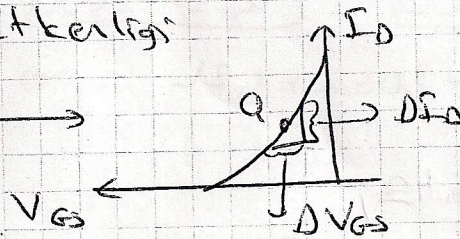
$$r_d = \frac{1}{y_{os}}$$

y_{os} : çıkış iletkenliği

g_m : FET geçiş iletkenliği

(EĞİM)

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}}$$



$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p}\right)^2$$

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{d}{dV_{GS}} I_D \Rightarrow$$

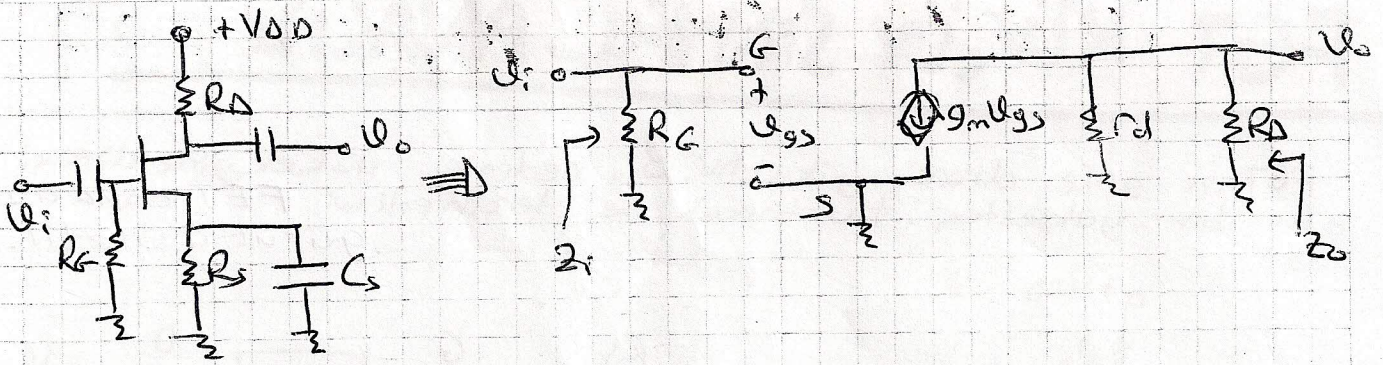
$$g_m = \frac{2I_{DSS}}{|V_p|} \left[1 - \frac{V_{GS}}{V_p}\right]$$

$$g_{m0} = \frac{2I_{DSS}}{|V_p|}$$

örnek üzere;

$$g_m = g_{m0} \left(1 - \frac{V_{GS}}{V_p}\right)$$

1. KENDİLİĞİNDEN KUTUPLANMIŞ DEVRE



1. DC analiz ile g_m bulunur.

2. $Z_i = R_G$ $Z_o = r_d || R_D$

3. $U_o = -g_m U_{gs} \cdot (r_d || R_D)$ $U_i = U_{gs} \Rightarrow A_v = -g_m (r_d || R_D)$

Örnek:

$V_{DD} = 16V$, $R_D = 1,2k$, $R_S = 200\Omega$; $I_{DSS} = 8mA$, $R_G = 1M\Omega$
 $V_p = -4V$, h_e $Z_i, Z_o, A_v = ?$

DC analiz:

$$\left. \begin{aligned} V_{GS} &= -R_S I_D \\ I_D &= I_{DSS} \left(1 - \frac{V_{GS}}{V_p}\right)^2 \end{aligned} \right\} I_D = 8m \left(1 + \frac{R_S I_D}{-4}\right)^2$$

Q çalışma noktası:

$V_{GSQ} = -0,94V$, $I_{DQ} = 4,7mA$ olarak bulunur.

Burada $g_m = 3,06m \frac{A}{V}$ bulunur. ($g_m = 3,06mS$)

$Z_i = 1M\Omega$

$Z_o = 1,2k\Omega$

$A_v = -g_m R_D$

$A_v = -3,06m \times 1,2k$

$\Rightarrow A_v = -3,67$

DC Analiz;

$V_{GS} = -R_S \cdot I_D = -(0,2k) \cdot I_D$ --- (1)

$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p}\right)^2 = 8 \left(1 - \frac{V_{GS}}{-4}\right)^2$ --- (2)

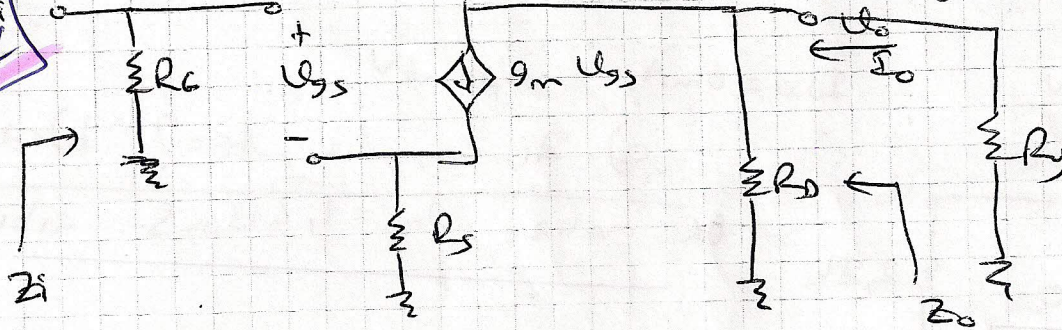
(1) ve (2) den $V_{GSQ} = -0,94V$ $I_{DQ} = 4,7mA$

$g_m = \frac{2 I_{DSS}}{|V_p|} \left(1 - \frac{V_{GSQ}}{V_p}\right) = \frac{2 \cdot 8}{4} \left(1 - \frac{(-0,94)}{(-4)}\right) = 3,06m \frac{A}{V}$
 $= 3,06mS$

2. SOURCE DİRENKLİ YÜKSELTEÇ

(Source direnci köprülenmiş)
- C_s yok

I_d YOK



$$Z_i = R_G$$

$$Z_o = R_D$$

U_i kısa devre olursa çevreden $U_{GS} + R_S - g_m U_{GS} = 0$ olur ve $U_{GS} = 0$ den sağlanır. yani akım kaynağı açık devre

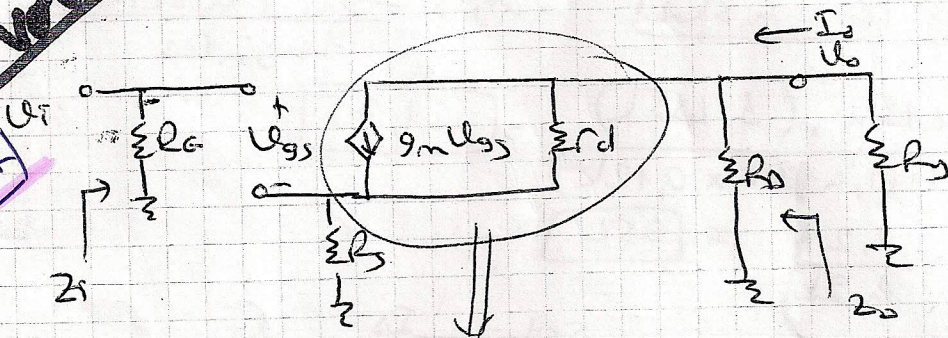
$$A_v = \frac{U_o}{U_i}$$

$$U_o = -g_m U_{GS} \cdot (R_D \parallel R_L)$$

$$U_i = U_{GS} + R_S \cdot (g_m U_{GS})$$

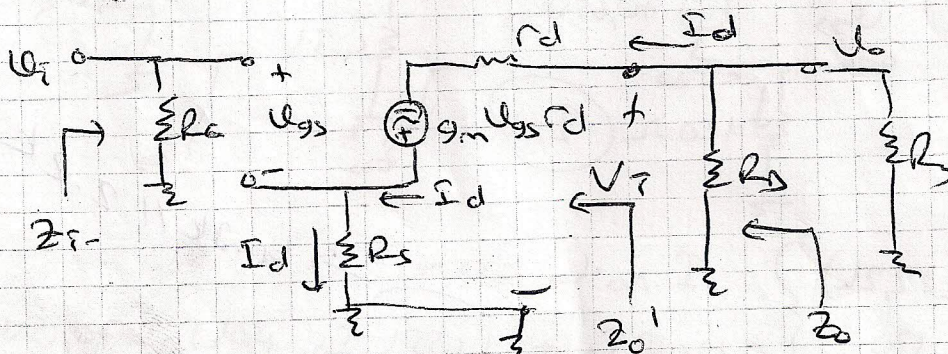
$$A_v = - \frac{g_m (R_D \parallel R_L)}{1 + g_m R_S}$$

I_d VAR



$$Z_i = R_G$$

kaynak döngüsüne ile:



$$U_o = -I_d \cdot (R_D \parallel R_L)$$

$$U_i = U_{GS} + I_d \cdot R_S$$

$I_d = \dots V_{GS}$ üzerinden yazılmalı. $\Rightarrow$ çıkışta çevre.

$$A_v = - \frac{g_m r_d (R_D \parallel R_L)}{g_m r_d R_S + r_d + R_S + (R_D \parallel R_L)}$$

Z_o için Z_o' bulalım

$$Z_o' = \frac{V_T}{I_d}$$

$V_{GS} = \dots I_d$ üzerinden bulunmalı. (buzutlar girişte çevreye girer diğer)

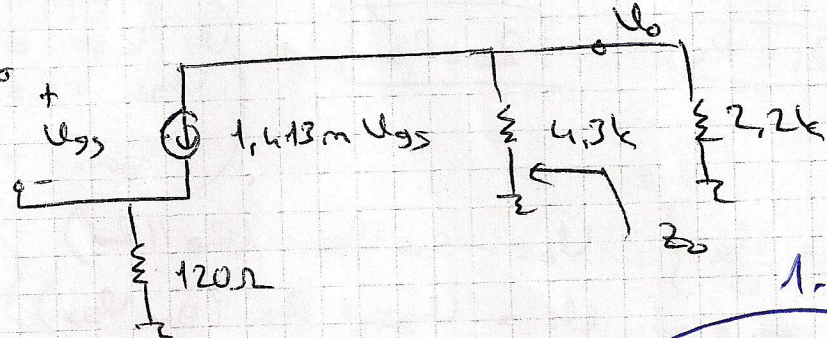
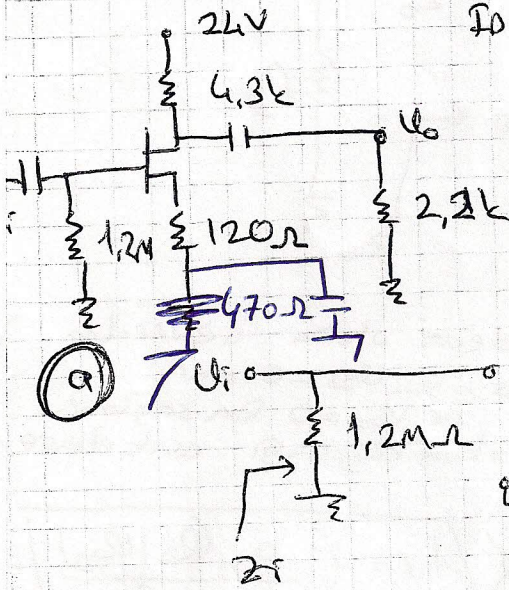
$$Z_o = r_d + R_S + g_m r_d R_S$$

Örnek:

$I_{DSS} = 6 \text{ mA}$, $V_p = -6 \text{ V}$

a) Z_i , Z_o , A_v , $A_r = ?$ b) $r_d = 50 \text{ k}$
 $r_{eac} = ?$

DC analiz: $g_m = 1,413 \text{ mS}$ çıkıyor.



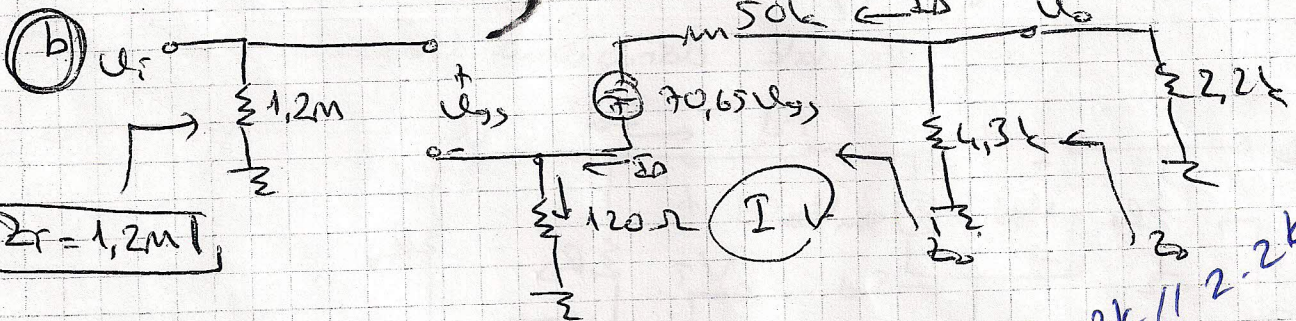
$Z_i = 1,2 \text{ M}\Omega$

$Z_o = 4,3 \text{ k}\Omega$

$A_v = \frac{-g_m (4,3 \text{ k}\Omega \parallel 2,2 \text{ k}\Omega)}{1 + g_m R_s}$

$A_v = \frac{-1,413 \text{ mS} (1,455 \text{ k}\Omega)}{1 + (1,413 \text{ mS} \times 0,12 \text{ k}\Omega)} = -1,75$

$A_r = +1,75 \cdot \frac{1,2 \text{ M}\Omega}{2,2 \text{ k}\Omega} = 954$



$Z_i = 1,2 \text{ M}\Omega$

$V_o = -I_D (4,3 \text{ k}\Omega \parallel 2,2 \text{ k}\Omega)$
 $V_r = V_{gs} + I_D \cdot 120$
 $0,120 \text{ k}\Omega I_D + 1,455 \text{ k}\Omega I_D + 50 \text{ k}\Omega I_D + 70,65 V_{gs} = 0$
 $V_{gs} = + \frac{51,575 \text{ k}\Omega I_D}{70,65} \Rightarrow V_{gs} = + 730 I_D$

$\Rightarrow V_o = -I_D \cdot 1,455 \text{ k}\Omega$
 $V_r = 850 I_D$
 $A_v = -1,711$
 $A_r = +1,711 \cdot \frac{1,2 \text{ M}\Omega}{2,2 \text{ k}\Omega}$

$A_r = 933,7$

$V_{gs} = 730 I_D$
yazılıyor

$Z_o \rightarrow$

Q

DC Analiz

$$V_{GSQ} = - (0.12 + 0.47) \cdot k \cdot I_D \quad \text{--- (1)}$$

$$I_D = I_{DSS} \left(1 - \frac{V_{GSQ}}{V_P} \right)^2$$

$$I_D = 6 \left(1 - \frac{V_{GS}}{-6} \right)^2 \quad \text{--- (2)}$$

(1) ve (2)'den; $V_{GSQ} = -1.76 \text{ V}$ bulunur.

$$g_m = \frac{2 I_{DSS}}{|V_P|} \cdot \left(1 - \frac{V_{GSQ}}{V_P} \right) = \frac{2 \cdot 6}{|-6|} \cdot \left(1 - \frac{-1.76}{-6} \right)$$

$$g_m = 1.413 \text{ mS}$$

FET $\leftrightarrow$ 4. sayfadaki ÖRNEK için

$$Z_{o1} = \frac{V_r}{I_D}$$

$$V_r = 50kI_D - 70,65V_{GS} + 120I_D$$

$V_r = 0$ için girer çıkarır:

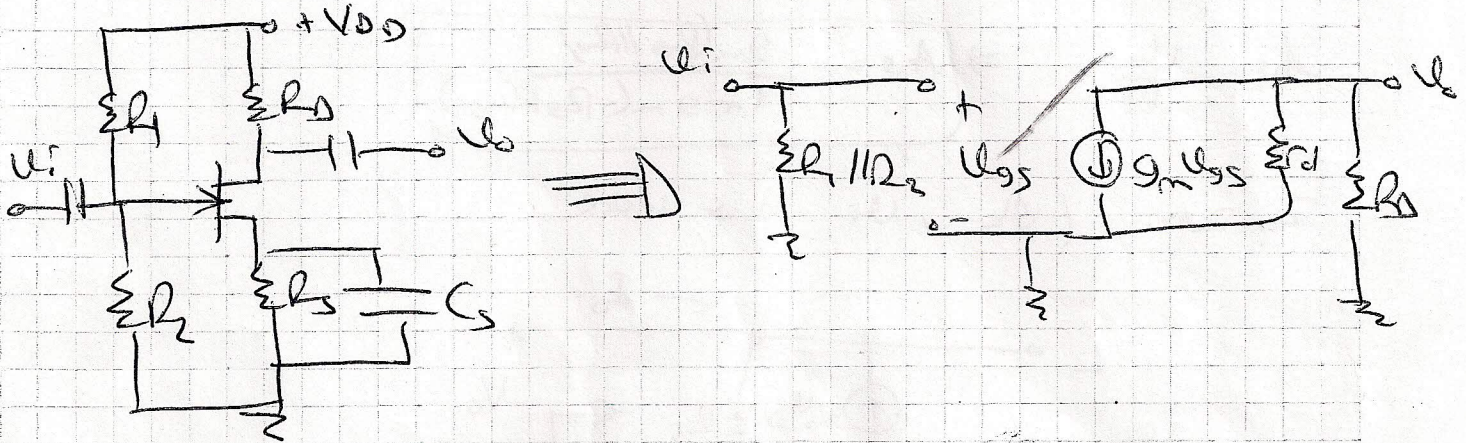
$$V_{GS} + 120I_D = 0 \Rightarrow V_{GS} = -120I_D$$

$$\Rightarrow V_r = 50kI_D + 8,478kI_D + 0,12kI_D$$

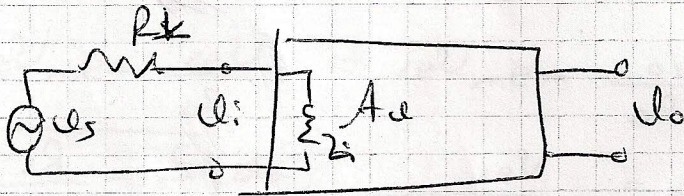
$$\Rightarrow Z_{o1} = 58,598k\Omega \approx 58,6k\Omega$$

$$Z_o = 4,3k\Omega // 58,6k\Omega = \underline{4k\Omega}$$

Voltaj Bölücü Konfigürasyon:



Kaynak Direncinin etkisi:

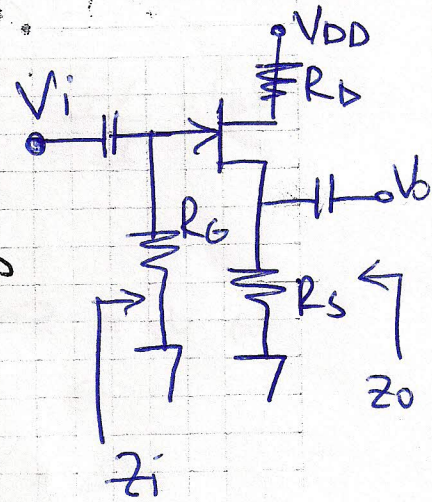
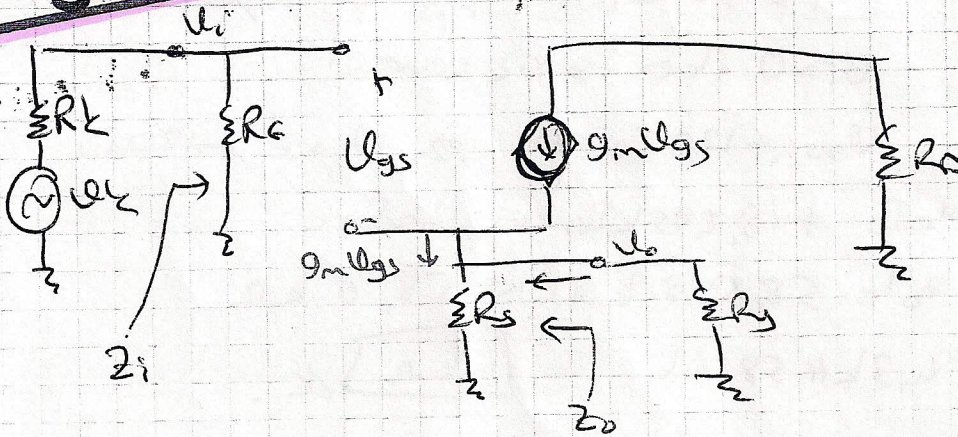


$$A_{vs} = A_e \cdot \frac{Z_i}{Z_i + R_k}$$

(örnekler yap
sayfa 25'le)

3) SOURCE İZLEYİCİ (ORTAK DRAIN) YÜKSELİCİ:

r_d yokken:

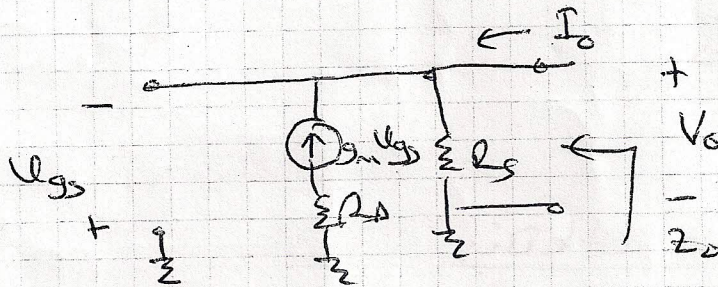


$$Z_i = R_G$$

$$u_o = g_m u_{gs} \cdot (R_S \parallel R_L), \quad u_i = u_{gs} + g_m u_{gs} (R_S \parallel R_L)$$

$$A_v = \frac{u_o}{u_i} \Rightarrow A_v = \frac{g_m (R_S \parallel R_L)}{1 + g_m (R_S \parallel R_L)}$$

Z_o izm: (u_k , u_i kısa devre.)



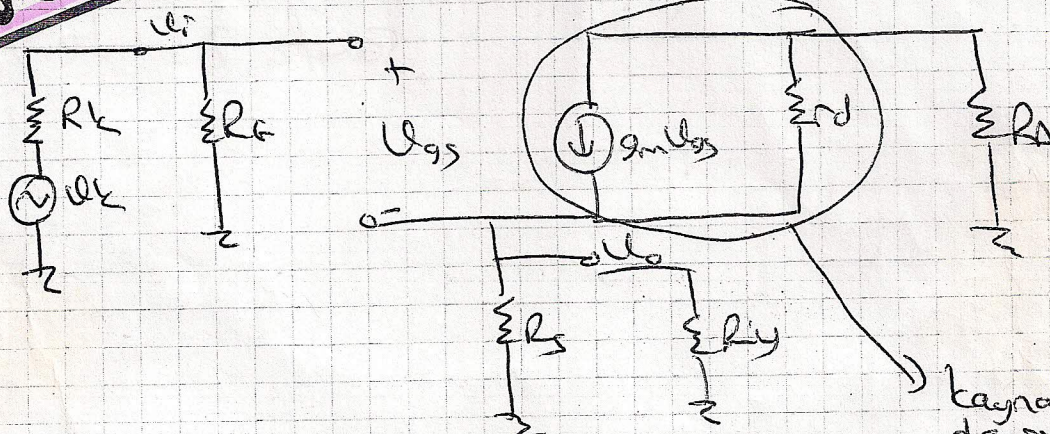
$$I_o = -V_{gs} \left(\frac{1}{R_S} + g_m \right)$$

$$V_o = -u_{gs}$$

$$I_o = \frac{V_o}{R_S} - g_m u_{gs} = -\frac{u_{gs}}{R_S} - g_m u_{gs}$$

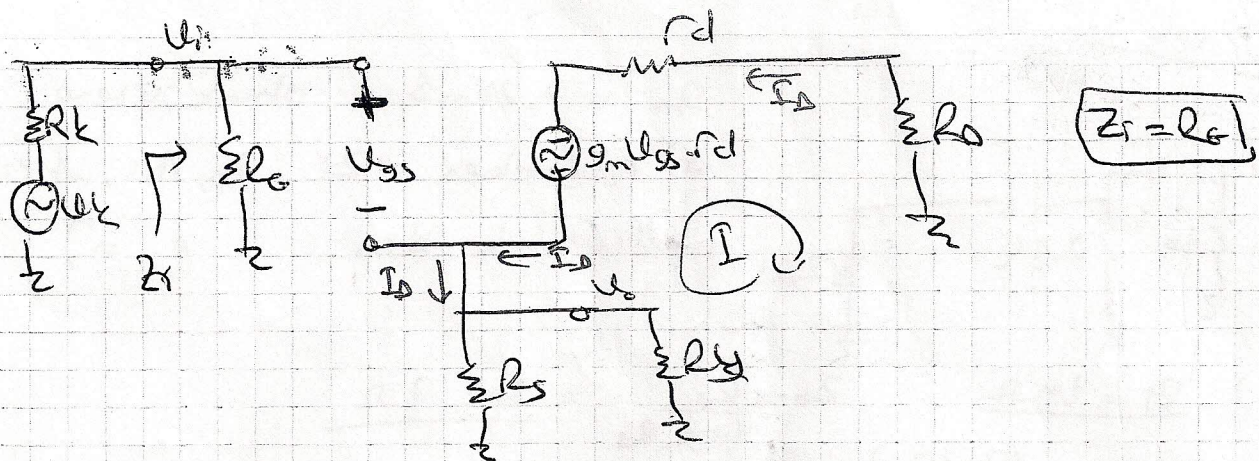
$$Z_o = V_o / I_o \Rightarrow Z_o = \frac{V_{gs}}{\frac{u_{gs}}{R_S} + g_m u_{gs}} \Rightarrow Z_o = \frac{R_S}{1 + g_m R_S}$$

r_d varken:



kaynak devresini

6



$$\left. \begin{aligned} U_o &= I_D \cdot (R_L \parallel R_D) \\ U_i &= U_{gs} + I_D (R_S \parallel R_L) \end{aligned} \right\} \begin{aligned} I_D &= U_{gs} \text{ üzerinden yararlanabiliriz.} \\ \text{çünkü } I_D &\text{ aynıdır.} \end{aligned}$$

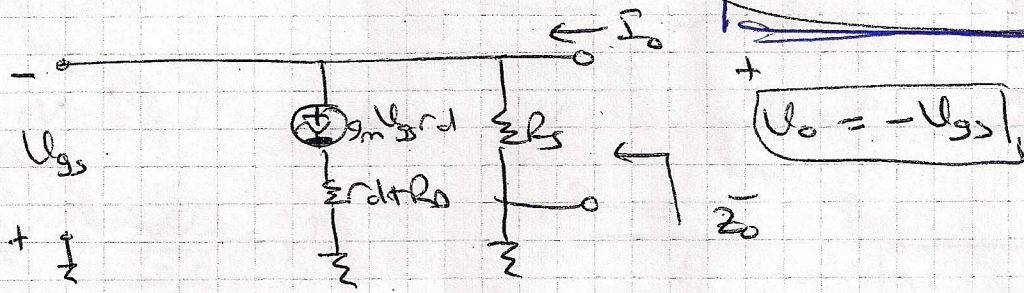
$$I_D (r_d + R_D) - g_m U_{gs} r_d + I_D (R_S \parallel R_L) = 0$$

$$I_D = \frac{g_m r_d}{(r_d + R_D) + (R_S \parallel R_L)} U_{gs}$$

$$\Rightarrow A_v = \frac{\frac{g_m r_d}{(r_d + R_D) + (R_S \parallel R_L)} \cdot (R_S \parallel R_L)}{1 + \frac{g_m r_d (R_S \parallel R_L)}{(r_d + R_D) + (R_S \parallel R_L)}}$$

$$\Rightarrow A_v = \frac{g_m r_d (R_S \parallel R_L)}{(r_d + R_D) + (R_S \parallel R_L) + g_m r_d (R_S \parallel R_L)}$$

$$A_v = \frac{g_m (r_d \parallel R_S \parallel R_L)}{1 + g_m (r_d \parallel R_S \parallel R_L)}$$



$$U_o = -U_{gs}$$

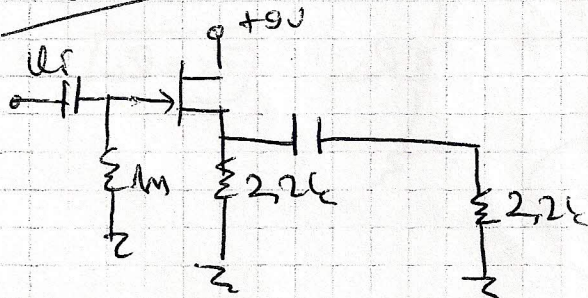
$$I_0 = -\frac{U_o}{R_S} + \frac{U_o - (+g_m U_{gs} r_d)}{r_d + R_D} = -\frac{U_{gs}}{R_S} - \frac{U_{gs} + g_m U_{gs} r_d}{r_d + R_D}$$

$$I_0 = -U_{gs} \left(\frac{r_d + R_D + R_S (1 + g_m r_d)}{r_d + R_D} \right)$$

$$Z_o = \frac{U_o}{I_0} \Rightarrow Z_o = \frac{r_d + R_D}{r_d + R_D + R_S + g_m r_d R_S}$$

$$Z_o = r_d \parallel R_S \parallel \frac{1}{g_m}$$

Örnek =



$g_m = 2,28 \text{ mS}$ ölmek üzere

a) rd yokken Z_i, Z_o, A_v, A_{v_i}

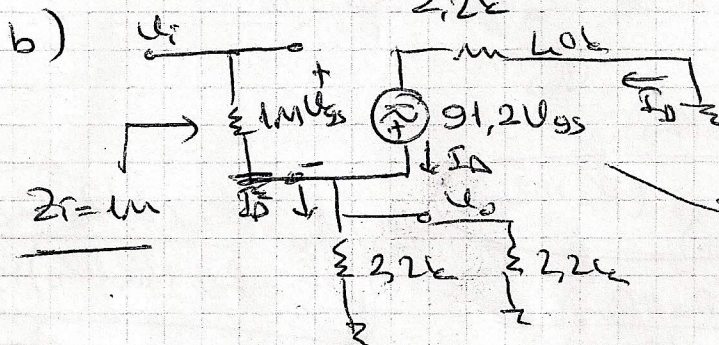
b) rd = 40k den $A_v = ?$

a) $Z_i = 1 \text{ M}\Omega$

$Z_o = \frac{R_s}{1 + g_m R_s} = 365,7 \Omega$

$A_v = \frac{g_m (R_s \parallel R_d)}{1 + g_m (R_s \parallel R_d)} = \frac{2,28 \text{ m} \cdot 1,1 \text{ k}}{1 + 2,28 \text{ m} \cdot 1,1 \text{ k}} = \frac{2,508}{3,508} = 0,714$

$A_r = -0,714 \cdot \frac{1 \text{ M}}{2,2 \text{ k}} \approx -325$



$40 \text{ k} I_D - 91,2 U_{gs} + 1,1 \text{ k} I_D = 0$

$I_D = \frac{91,2}{41,1 \text{ k}} U_{gs}$

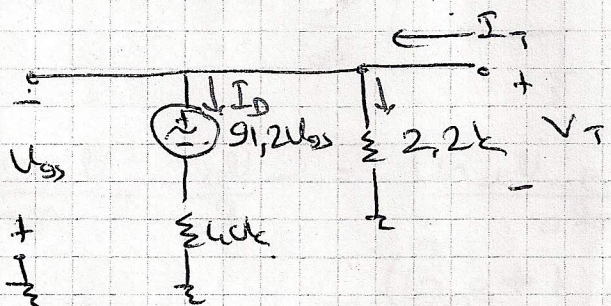
$I_D = 2,219 \text{ m} U_{gs} / \text{k}$

$U_o = 2,219 \text{ m} U_{gs} \cdot 1,1 \text{ k} = 2,44 U_{gs}$

$A_v = \frac{2,44}{3,44} = 0,71$

$U_i = U_{gs} + U_o$
 $U_i = 3,44 U_{gs}$

$Z_o = V_T / I_T$ $V_T = -U_{gs}$ $I_T = \frac{V_T}{2,2 \text{ k}} + I_D$



$I_T = \frac{-U_{gs}}{2,2 \text{ k}} + \frac{V_T - 91,2 U_{gs}}{40 \text{ k}}$

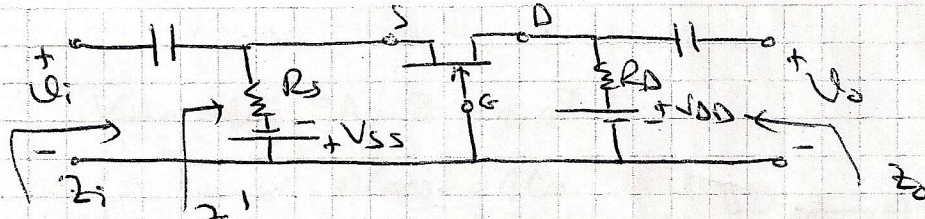
$I_T = \frac{-U_{gs}}{2,2 \text{ k}} - \frac{U_{gs} - 91,2 U_{gs}}{40 \text{ k}}$

$I_T = -U_{gs} \left(\frac{1}{2,2 \text{ k}} + \frac{92,2}{40 \text{ k}} \right)$

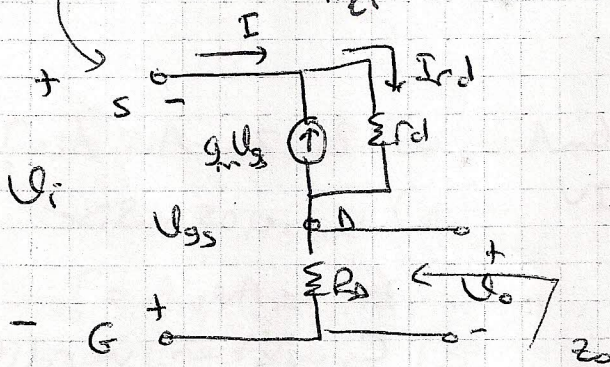
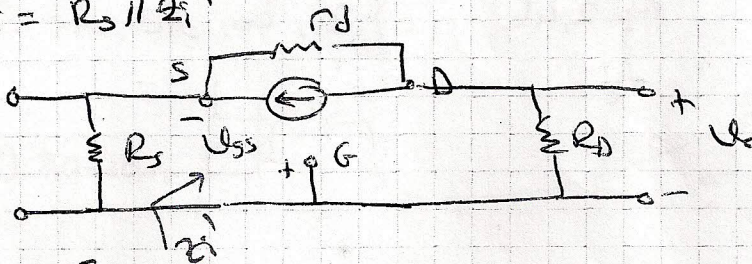
$Z_o = \frac{88 \text{ M}}{242,8 \text{ k}} = 362,38 \Omega$

veya $Z_o = r_d \parallel R_s \parallel \frac{1}{g_m}$ der

4. ORTAK GATE KONFIGÜRASYON:



$$Z_i = R_S \parallel Z_i'$$



$$V_{rd} + V_{R_D} = U_i$$

$$V_{rd} = U_i - I \cdot R_D$$

$$V_{rd} = I_{rd} \cdot r_d = (I + g_m U_{gs}) \cdot r_d$$

$$I \cdot r_d + g_m U_{gs} \cdot r_d = U_i - I \cdot R_D$$

$$Z_i' = \frac{U_i}{I} \Rightarrow I (r_d + R_D) = U_i - g_m U_{gs} r_d$$

$$U_{gs} = -U_i \quad \text{old. form} \Rightarrow$$

$$I = \frac{U_i (1 + g_m r_d)}{r_d + R_D} \Rightarrow Z_i' = \frac{r_d + R_D}{1 + g_m r_d}$$

$$Z_i = R_S \parallel \left(\frac{r_d + R_D}{1 + g_m r_d} \right)$$

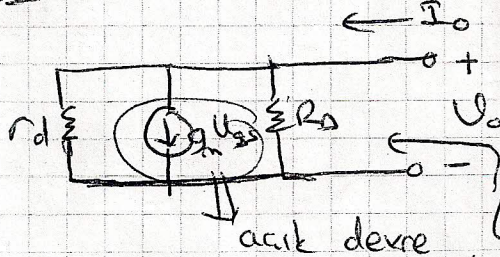
$$A_v = \frac{U_o}{U_i} \Rightarrow$$

$$U_o = I \cdot R_D$$

$$U_o = \frac{U_i \cdot (1 + g_m r_d)}{r_d + R_D} \cdot R_D$$

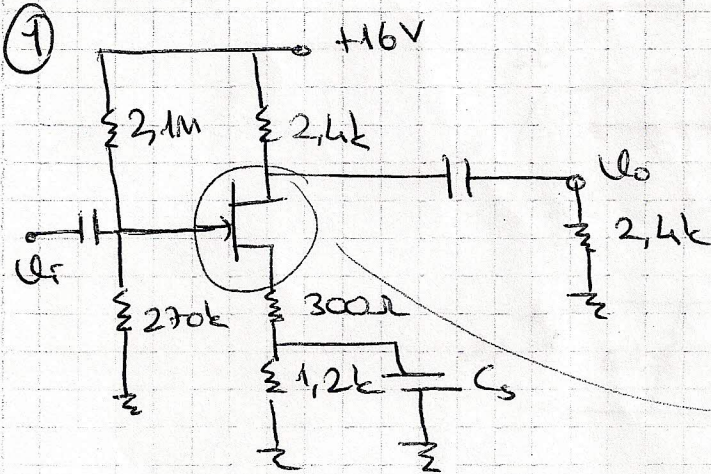
$$A_v = \frac{(1 + g_m r_d) \cdot R_D}{r_d + R_D}$$

$$Z_o: U_i = 0.$$



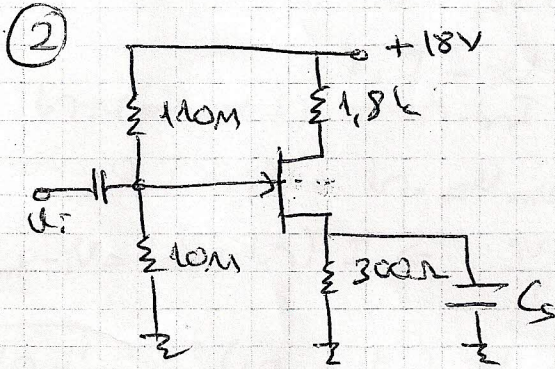
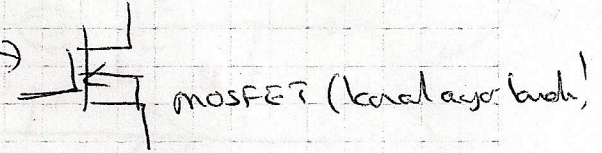
$$Z_o = r_d \parallel R_D$$

Örnek:



$I_{DSS} = 8\text{mA}$ $V_p = -4\text{V}$

- a) DC analiz $g_m = ?$
 b) AC eşdeğer devre çiz.
 c) $Z_i, Z_o, A_v, A_i = ?$



$I_{DSS} = 6\text{mA}$
 $V_p = -3\text{V}$

- a) $Z_i, Z_o, A_v, A_i = ?$
 b) $y_{os} = 0,05\text{mS}$
 $Z_i, Z_o, A_v, A_i = ?$
 c) C_s yokken verditimel
 $Z_i, Z_o, A_v, A_i = ?$
 d) C_s yokken $y_{os} = 0,10\text{mS}$
 Be $A_v, A_i, Z_i, Z_o = ?$